

Definition. Suppose the function $f(x)$ is defined for x in the interval $[a, b]$. Then a **Riemann sum** for $f(x)$ on $[a, b]$ is an expression of the form

$$f(x_1) \cdot \Delta x_1 + f(x_2) \cdot \Delta x_2 + \cdots + f(x_n) \cdot \Delta x_n.$$

The interval $[a, b]$ has been divided into n subintervals whose lengths are $\Delta x_1, \dots, \Delta x_n$, respectively, and for each k from 1 to n , x_k is some point in the k -th subinterval.